Seungeun Lee

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https://duneag2.github.io

Research Areas

Machine Learning, Explainable AI, Trustworthy AI; Enumerative Combinatorics

Education

New York University — New York, USA
Ph.D. Student in Computer Science — Sep 2025 - Present
- Advisor: Prof. Julia Stoyanovich

Korea University — Seoul, Korea
M.S. in Mathematics (Applied Mathematics) — Mar 2024 - Aug 2025
- Advisor: Prof. Seungsang Oh
- GPA: 4.00/4.00 (Integrated B.S./M.S. Program)

Korea University — Seoul, Korea
B.S. in Mathematics, B.E. in Computer Science and Engineering — Mar 2020 - Feb 2024
- GPA: 3.76/4.00, Great Honor (Magna Cum Laude)

Publications

1. H. Hwang, **S. Lee**, L. Rosenblatt, S. E. Whang, J. Stoyanovich, *Explanation Multiplicity in SHAP: Characterization and Assessment*, NeurIPS 2026
2. **S. Lee**, S. Choi, T. Kang, S. Chung, S. Han, J. Seo, S. Oh, and I.-Y. Kwak, *iWAX: Interpretable Wav2vec-AASIST-XGBoost Framework for Voice Spoofing Detection,* Scientific Reports
3. S. Chung, H. Kim, **S. Lee**, S. Lee, and S. Oh, *Honeycomb-lattice monomer-dimer mixtures,* Journal of Statistical Physics
4. J. Kim*, **S. Lee***, S.M. Ahn, G. Choi, B.-K. Je, B.J. Park, Y. Cho, and S. Oh, *Deep Learning Model Using Cross-sequence Learning to Identify Orbital Fractures in Radiographs of Patients Under 20 Years,* Frontiers in Bioengineering and Biotechnology (*: equal contribution)
5. S. Han*, J. Seo*, S. Choi, T. Kang, S. Chung, **S. Lee**, S. Park, S. Oh, and I.-Y. Kwak, *Enhancing voice spoofing detection in noisy environments using frequency feature masking augmentation,* Engineering Science and Technology, an International Journal (*: equal contribution)
6. **S. Lee***, N. Jeong*, J. Je*, and S.-Y. Lee, *BCCP: An MLOps Framework for Self-cleansing Real-Time Data Noise via Bayesian Cut-off-based Closest Pair Sampling*, IEEE International Conference on Artificial Intelligence x Data & Knowledge Engineering 2024 (*: equal contribution)
7. S. Choi, S. Chung, **S. Lee**, S. Han, T. Kang, J. Seo, I.-Y. Kwak, and S. Oh, *TB-ResNet: Bridging the gap from TDNN to ResNet in Speaker Verification with Temporal-Bottleneck Enhancement,* ICASSP 2024
8. **S. Lee***, Y. Cho*, Y. Ji, M. Jeon, A. Kim, B.-J. Ham*, and Y.Y. Joo*, *Multimodal integration of neuroimaging and genetic data for the diagnosis of mood disorders based on computer vision models,* Journal of Psychiatric Research (*: equal contribution)
9. **S. Lee**, S.H. Hwang, S. Oh, B.J. Park, and Y. Cho, *Multi-input Vision Transformer with Similarity Matching,* MICCAI PRIME Workshop 2023

Preprints

1. **S. Lee**, R. Mukhopadhyay, A. Bakshi, J. Stoyanovich, *MissSketch: Characterizing and Mitigating Missingness-Induced Sketch Amplification*
2. **S. Lee**, J. Fonseca, J. Stoyanovich, *RelShap: Relationally Consistent Shapley Explanations*

3. **S. Lee**, K. Lee, S. Bae, S. Lee, S. Lee, J. Stoyanovich, I.-Y. Kwak, and S. Oh, *Table2Image: Lightweight Tabular Learning with Generated Proxy Representations and Reliability Diagnostics*
4. S. Choi, **S. Lee**, S. Oh, *Integral square dissections of Möbius bands*
5. **S. Lee**, N. Y. Han, G. Yang, and Y. Cho, *CurvUNet: Curvature-Informed UNet for Enhanced Medical Image Segmentation*
6. **S. Lee**, S. Chung, S. Lee, and S. Oh, *Counting perfect matchings in torus grid graphs*
7. T. Kang, S. Han, S. Choi, J. Seo, S. Chung, **S. Lee**, S. Oh and I.-Y. Kwak, *Experimental Study: Enhancing Voice Spoofing Detection Models with wav2vec 2.0*

Research Experience

Center for Responsible AI, New York University — New York, USA
Graduate Research Assistant — Sep 2025 - Present
- Advisor: Prof. Julia Stoyanovich

DLmath Lab, Korea University — Seoul, Korea
Undergraduate Research Intern, Graduate Research Assistant — Apr 2023 - Aug 2025
- Advisor: Prof. Seungsang Oh

Department of Radiology, Korea University Anam Hospital — Seoul, Korea
Undergraduate Research Intern — Apr 2022 - Mar 2023
- Advisor: Prof. Yongwon Cho

Institute of Data Science, Korea University — Seoul, Korea
Undergraduate Research Intern — Nov 2021 - Mar 2022
- Advisors: Prof. Yoonjung Yoonie Joo and Prof. Yongwon Cho

Work Experience

SCIEN — Ulsan, Korea
Data Scientist & Engineer Intern — Jul 2024 - Aug 2024
- Developed pill efficacy ML/DL models with Handok Pharmaceuticals

Photorage — Seoul, Korea
Technical Advisor — Sep 2023
- Provided technical guidance on image generation and recommendation systems for a photo storage app at a startup

Teaching Experience

New York University — New York, USA
Section Leader and Course Assistant — Fall 2026
- Responsible Data Science (DS-UA 202)

Korea University — Seoul, Korea
Teaching Assistant — Spring 2024, Fall 2024, Spring 2025
- Topology I (MATH 333), Deep Learning Math (MATH 391)

Korea Human Resource Development Institute for Health & Welfare — Seoul, Korea
Teaching Assistant — Aug 2022 - Nov 2022
- Guided a medical AI project on CT-free orthopedic diagnosis using DRR and X-ray segmentation. Evaluated final competition entries.

Mentoring Experience

Seunghyun Lee — Mar 2025 - Feb 2026
Undergraduate mentee (Co-supervised with Prof. Seungsang Oh); project on suppressive dropout.

Awards and Honors

- OpenMined Fellowship (Inaugural) — Sep 2026 - Feb 2027
- Dissertation Fellowship, New York University (NYU) — Jun 2026 - Aug 2026
- School of Engineering Fellowship, NYU — Sep 2025 - Aug 2026
- Teaching & Research Assistant Scholarship, Korea University (KU) — Mar 2024 - Aug 2025
- Scholarship for Integrated B.S./M.S. Program Students, KU — Mar 2024 - Aug 2025
- Scholarship for Veritas Program (REU, College of Informatics, KU) — Fall 2023
- Scholarship for Veritas Program (REU, College of Science, KU) — 2023
- Samsung SDS Scholarship — 2020 - 2023
- Dean's List (College of Science, KU) — Fall 2020
- Semester High Honors, KU — 2020 - 2023

Languages

Korean (native), English (fluent), Spanish (basic), Chinese (basic)